

Saksham Saxena

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EDUCATIONAL QUALIFICATIONS

Bachelor of Technology in Computer Science, GLA University, Mathura
Intermediate, Romex International School
High School, Romex International School

May 2026
March 2022
March 2020

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, Javascript
- **AI/ML:** Machine Learning, Deep Learning, Large Language Models (LLMs), Transformers, RAG Pipelines, Prompt Engineering, NLP.
- **Frameworks & Databases:** FastAPI, Flask, React.js, Streamlit, MongoDB, ChromaDB (Vector DB)
- **Developer Tools:** Git/GitHub, Docker, VS Code, Google Colab.

PROJECTS

LLM-Powered Real-Time Database Query Engine

- Architected an asynchronous **NL2Query (Text-to-NoSQL)** engine using **FastAPI** and **Streamlit** to translate natural language intent into complex, multi-stage **MongoDB aggregation pipelines**.
- Optimized context efficiency of **Groq-hosted Llama-3.1-8b** via dynamic schema-injection and few-shot prompting, achieving an execution latency of **< 250ms**.
- Engineered a zero-trust security framework featuring a custom **validation layer** and strict **RBAC**, completely mitigating prompt injection vulnerabilities and enforcing 100% read-only data access.

AI-powered Semantic Search System

- Engineered an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline using **Groq Llama-3** and **ChromaDB** to enable low-latency, context-aware question answering over unstructured enterprise data.
- Designed a robust multi-modal data ingestion pipeline using **PyMuPDF** and **Tesseract OCR**, implementing optimal text-chunking strategies to extract and index data from PDFs, images, and DOCX files
- Optimized vector similarity search execution by exposing specialized **RESTful APIs** via **Flask**, ensuring high-throughput semantic retrieval and seamless LLM context delivery.

Handwritten Equation Solver

- Built a hybrid AI system using **PyTorch**, **OpenCV**, and **SymPy** to parse, recognize, and evaluate complex mathematical expressions via transformer-based OCR and advanced image preprocessing.
- Developed high-performance dual-framework APIs using **FastAPI** and **Flask** to handle concurrent image uploads, real-time symbolic parsing, and step-by-step mathematical breakdowns.
- Implemented an intelligent **Google Gemini fallback routing mechanism** for low-confidence local OCR predictions, significantly boosting total recognition accuracy across diverse symbolic edge cases.

RESEARCH & PUBLICATIONS

- **Co-Author**, "*Predicting Telecom Churn Through Combined Machine Learning Models*" – Presented at **IEEE International Conference** (2025).

CERTIFICATIONS

- Machine Learning from Udemy
- OCI 2025 certified AI Foundations Associate.

ACHIEVEMENTS

- Top 15 in "The AI Hiring Show" Hackathon conducted by Rabbit Learning AI.
- Top 10 in "IBM ICE Day" hackathon Organised by IBM.

EXTRACURRICULAR ACTIVITIES

- President of NSS GLAU Mathura.
- Mentored 20 plus teams in SIH Internal Hackathon 2025.
- Guided 15 plus teams in GLA National Hackathon 2025 "*Deviathon*".